Cyber Security CTF Machine Development Report

Title:

Cyber security CTF Machine Development Assignment

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Course: Cyber Security

Operating System :kali Linux

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1. Objective

The objective of this assignment was to design and deploy a vulnerable Capture The Flag (CTF) machine on a Linux environment. The machine simulates a real-world penetration testing target where participants perform reconnaissance, enumeration, web exploitation, SSH access, privilege escalation, and steganography to capture flags. The machine is intentionally configured with hidden clues that require logical analysis instead of direct disclosure of credentials.

2. Machine Specifications

Specification Details

Virtualization Software: VMware Workstation
Development Environment: Kali Linux
Target Machine: Ubuntu 14.04.6 LTS
Web Server: Apache2
SSH Service: OpenSSH
Hostname: madtha
Firewall: UFW
Browser Tested: Firefox

3. Operating System Used

The operating system used for developing the CTF machine is **Kali Linux**, which provides various tools required for penetration testing and security analysis. The target machine was configured using **Ubuntu 14.04.6 LTS**, which served as the vulnerable system for the CTF challenge.

The combination of Kali Linux and Ubuntu created a realistic environment for performing reconnaissance, enumeration, exploitation, privilege escalation, and steganography-based challenges.

4. Network Configuration

The machine was configured using NAT networking allowing communication between the attacker and target systems.

Accessible ports:
Port 22 (SSH)
Port 80 (HTTP)

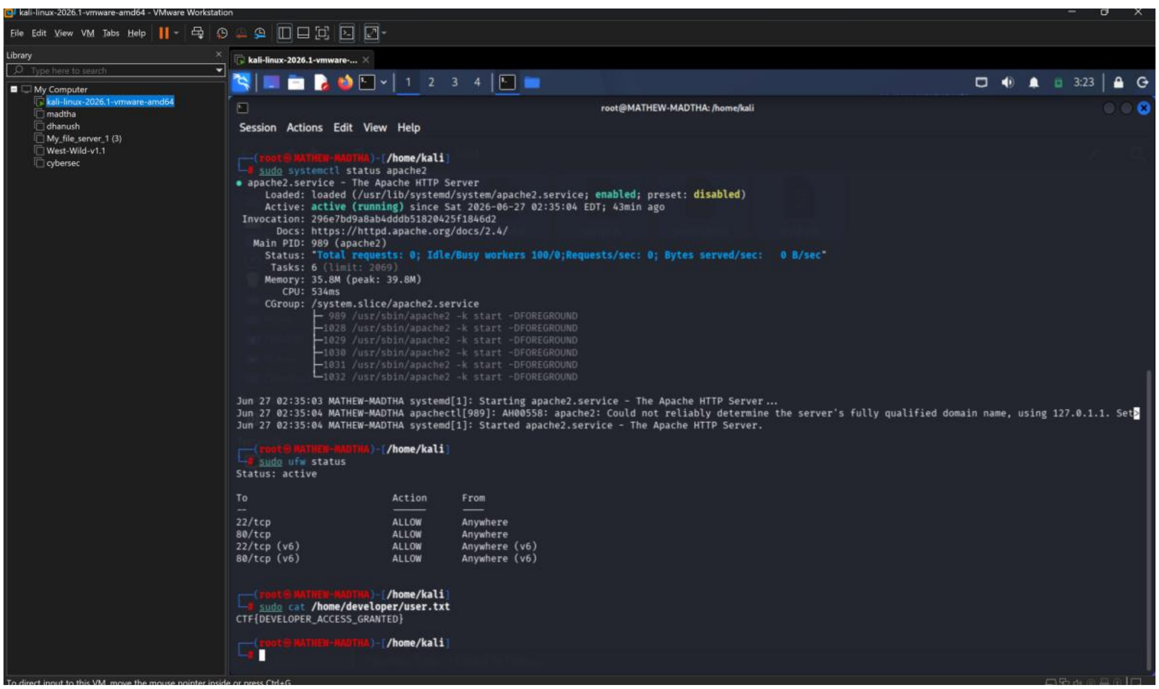
All unnecessary ports were blocked using the firewall.

5. User Accounts Created

Three separate Linux user accounts were created.

- mathew
- mokshith
- neha

Each user has an individual home directory and login shell. Credentials were hidden inside the challenge environment.



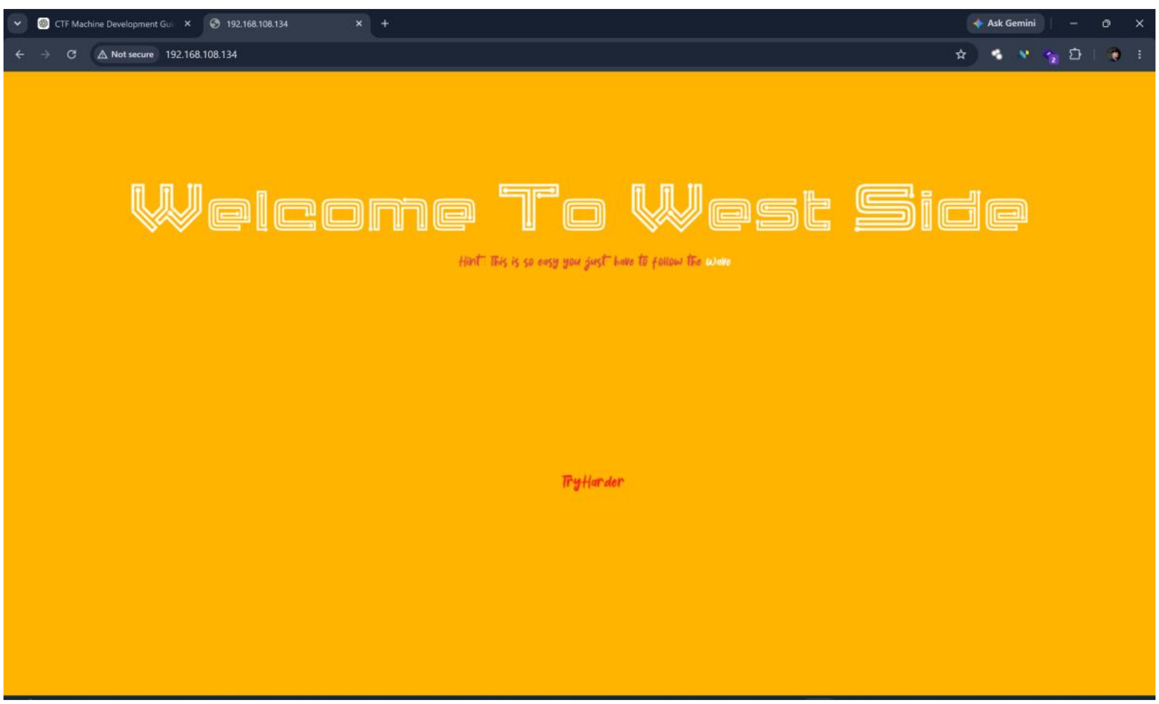
6. Services Configured

The following services were configured.

Apache Web Server
• Runs on Port 80
• Hosts portfolio website
• Contains hidden clues

SSH
• Runs on Port 22
• Used after discovering credentials

Firewall
• Configured using UFW allowing SSH and HTTP



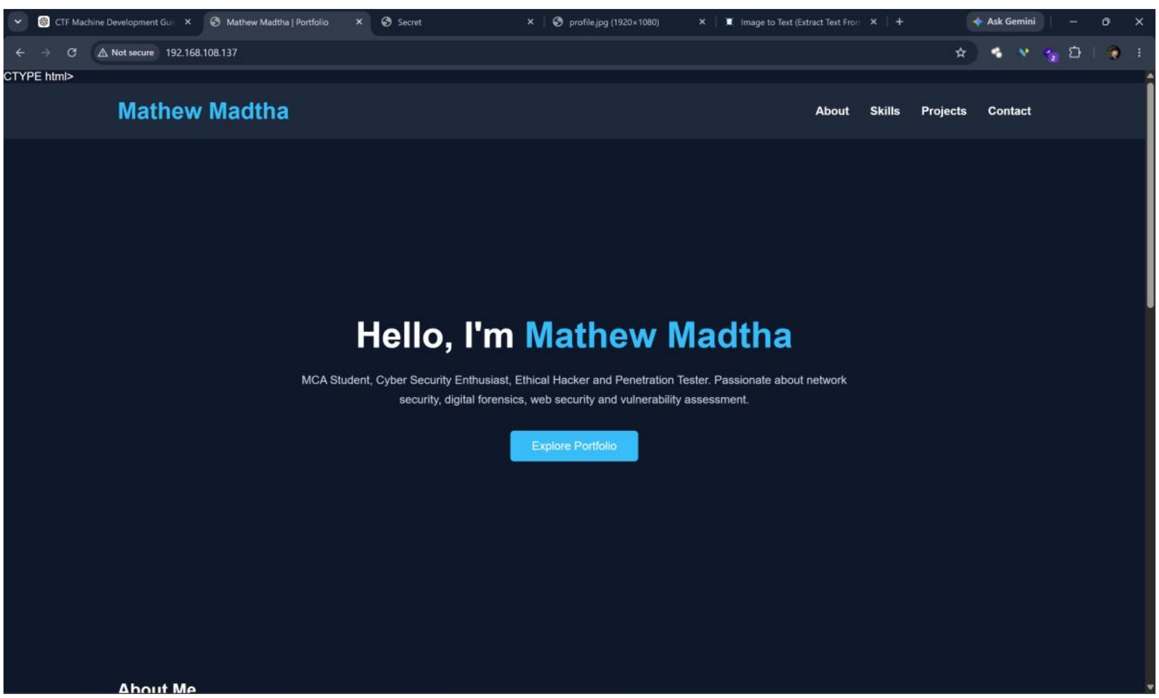
7. Website Design Overview

A personal portfolio website was developed containing:

• Professional profile
• Skills
• Projects
• Contact information

The website contains intentionally hidden information for participants to discover.

Website technologies:
HTML
CSS
JavaScript

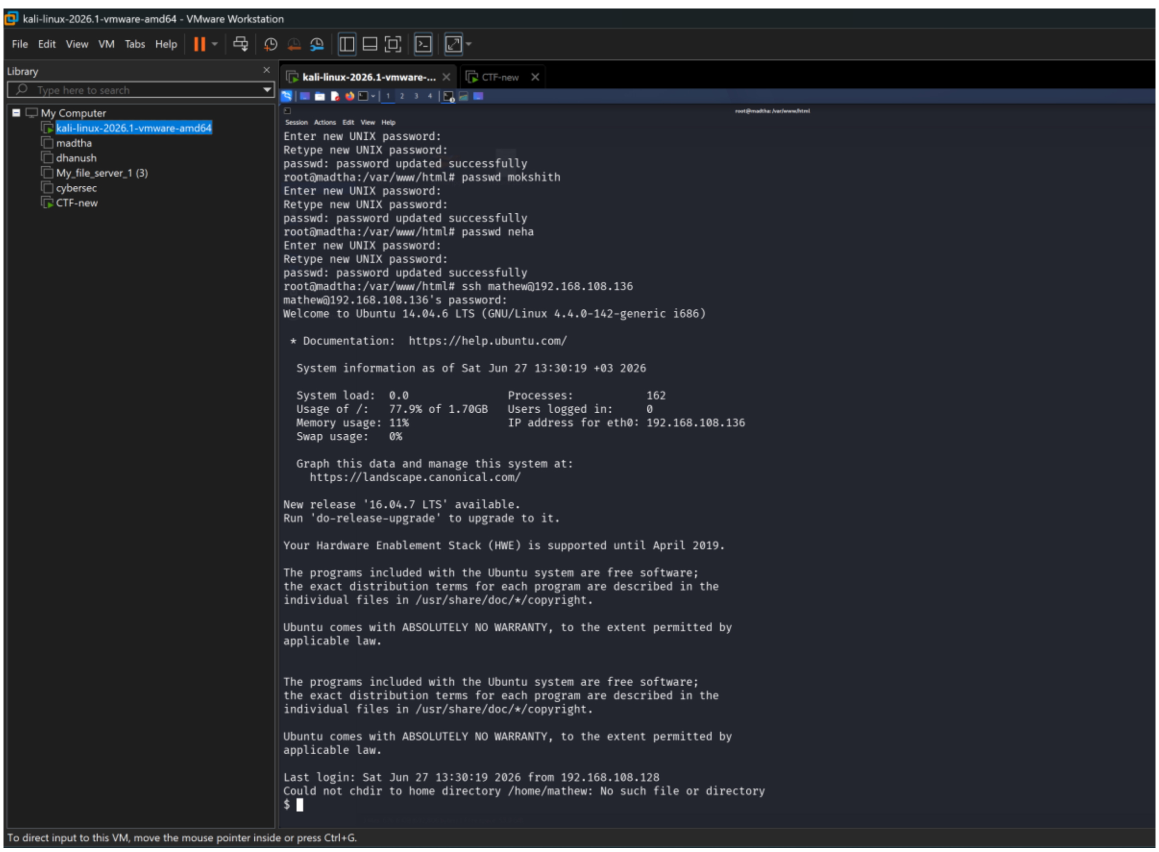


8. Hidden Credentials Methodology

The machine requires enumeration.

Hidden information was distributed using:

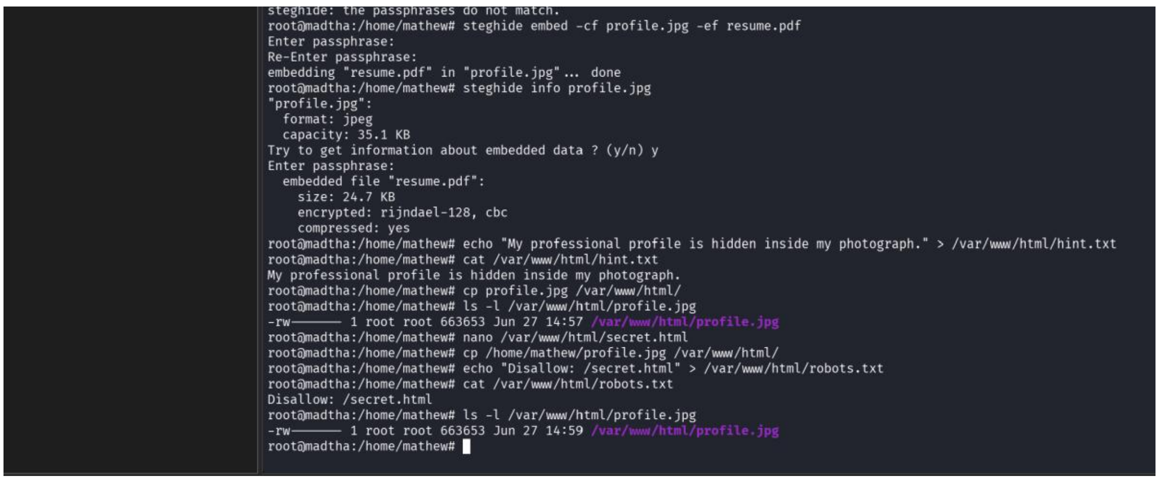
- HTML comments
- JavaScript source files
- CSS comments
- robots.txt
- Hidden directories
- Encoded strings (Base64)
- Source code inspection



9. Steganography Implementation

A file named profile.jpg was used for steganography. A professionally formatted resume.pdf was embedded into the image using steghide.

Example extraction:
steghide extract -sf profile.jpg



10. Flag Locations and Hostname

Two flags were created.

User Flag:
/home/wavex/wave/FLAG1.txt

Root Flag:
/root/FLAG2.txt

Hostname:
madtha

11.Screenshots of Configuration

Include the following screenshots:

- Hostname configuration
- User accounts
- Apache service status
- SSH service status
- Firewall configuration
- Website homepage
- robots.txt
- Hidden page
- Steganography information
- User flag
- Root flag

12. Hostname Configuration

The system hostname was configured as madtha to uniquely identify the target machine within the virtual network environment. The hostname configuration helps distinguish the CTF machine from other systems during reconnaissance and penetration testing activities.

The hostname was verified using Linux system commands and was used throughout the configuration and testing phases of the project. Assigning a unique hostname provides better system identification and management in a controlled cybersecurity environment.

Configured Hostname: madtha

13. Testing Procedure

Step 1: Port scanning using nmap
Step 2: Visit website
Step 3: Inspect source code
Step 4: Check robots.txt
Step 5: Enumerate hidden files
Step 6: Decode hidden strings
Step 7: Recover credentials
Step 8: SSH Login
Step 9: Capture user flag
Step 10: Privilege escalation
Step 11: Capture root flag
Step 12: Extract hidden resume

14. Vulnerability Walkthrough

1. Perform network scan to identify open services.
2. Access the hosted portfolio website.
3. Inspect HTML, CSS, JavaScript, and robots.txt for hidden clues.
4. Decode hints to discover credentials.
5. Authenticate via SSH.
6. Retrieve the user flag.
7. Escalate privileges.
8. Retrieve the root flag.
9. Extract the embedded resume using steganography.

15. Conclusion

This project demonstrates the design and deployment of a realistic Capture The Flag machine using Kali Linux. The machine integrates web enumeration, credential discovery, SSH authentication, privilege escalation, firewall configuration, and steganography into a single practical exercise.

Appendix

Software Used:

Kali Linux

Ubuntu 14.04.6 LTS

Apache2

OpenSSH

UFW Firewall

Steghide

VMware Workstation

Firefox Browser